Takuya Okawa

IFPU Postdoctoral Researcher

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RESEARCH INTERESTS

Astrophysical and cosmological probes of dark sector physics, with emphasis on dark matter, sterile neutrinos, and axions. The Dirac or Majorana nature of heavy neutral leptons. Dynamics of a dark-sector scalar in the early universe and its connection to the generation of primordial curvature perturbations.

EMPLOYMENT

IFPU Postdoctoral Researcher, Scuola Internazionale Superiore di Studi Avanzati (SISSA) Sep. 2025 – present

EDUCATION

Ph.D., Physics, Washington University in St. Louis Aug. 2025
Advisor: Francesc Ferrer, P. S. Bhupal Dev
B.S., Physics, University of Tokyo March 2020

EXPERIENCE

Visiting Scholar, Fermilab July – Dec. 2024, Sep. 2026
Visiting Scholar, CERN Nov. – Dec. 2025
Visiting Student, Princeton University Sep. – Nov. 2019

PUBLICATIONS

7. T. Okawa and S. T. Petcov, "Low-Scale See-Saw Mechanism and the Majorana Nature of Heavy Neutrinos," arXiv:2610.21886.
6. C. V. Cappiello, S. Das, P. S. B. Dev, T. Okawa, and S. Razzaque, "Reappraisal of the Constraints on Heavy Axion-like Particles from Gamma-Ray Bursts," arXiv:2607.07297.
5. J. H. Buckley, P. S. B. Dev, F. Ferrer, and T. Okawa, "Probing Heavy Axion-like Particles from Massive Stars with X-rays and Gamma Rays," arXiv:2412.21163.
4. A. Ireland, G. Krnjaic, and T. Okawa, "The Affleck-Dine Curvaton," Phys. Rev. D 113 (2026) 103537, arXiv:2410.13712.
3. S. Das, P. S. B. Dev, T. Okawa, and A. Soni, "Old neutron stars as a new probe of relic neutrinos and sterile neutrino dark matter," Phys. Rev. D 111 (2025) 055035, arXiv:2408.01484.
2. T. Okawa and A. Y. Chen, "A Model for Pair Production Limit Cycles in Pulsar Magnetospheres," Astrophys. J. 970 (2024) 46, arXiv:2402.19436.
1. P. S. B. Dev, F. Ferrer, and T. Okawa, "On the Galactic radio signal from stimulated decay of axion dark matter," JCAP 04 (2024) 045, arXiv:2311.13653.

AWARDS

Dean's award for Graduate Research Excellence, semifinalist Aug. 2024

URA visiting scholar award	April 2024
Pitt PACC travel award	May 2023
Ito Foundation U.S.A.-FUTI Scholarship	May 2020

TALKS

* Invited talk

*Boston University, Department of Physics**	Sep. 2026
*Pennsylvania State University, GAPP Seminar**	Sep. 2026
*Washington University in St. Louis, Physics Theory Seminar**	Sep. 2026
*University of Oklahoma, High Energy Physics Seminar**	Sep. 2026
*Majorana–Raychaudhuri Seminars (online)**	June 2026
TAsP Meeting 2026	Jan. 2026
CERN-TH Cosmo Coffee	Nov. 2025
TeVPA 2025	Nov. 2025
*BRDA 2025: Selected Topics in (B)SM**	Oct. 2025
SISSA–ICTP Joint Journal Club	Oct. 2025
*IFPU Friday Talk**	Oct. 2025
Invisibles25 Workshop	Sep. 2025
CETUP 2025*	June 2025
XVIII International Conference on Interconnections between Particle Physics and Cosmology	June 2025
Cosmology and astrophysics group meeting, WUSTL	June 2025
5th EuCAPT Annual Symposium (online)	May 2025
*University of Winnipeg Theory Seminar (online)**	Jan. 2025
*Argonne National Laboratory Theory Seminar**	Dec. 2024
*Virginia Tech Center for Neutrino Physics Seminar**	Nov. 2024
Particle Physics on the Plains 2024	Nov. 2024
Cosmology and astrophysics group meeting, WUSTL	Sep. 2024
TeV Particle Astrophysics 2024	Aug. 2024
TASI 2024	June 2024
DPF-PHENO 2024	May 2024
Computational Astro Theory group meeting, WUSTL	Feb. 2024
Cosmology and Astrophysics group meeting, WUSTL	Feb. 2024
Cosmology and Astrophysics group meeting, WUSTL	Dec. 2023
Particle Physics on the Plains 2023	Oct. 2023
Brookhaven Forum 2023 Advancing Searches for New Physics	Oct. 2023
N3AS Summer School 2023	July 2023
Phenomenology 2023 Symposium	May 2023
APS April meeting 2023	April 2023
Particle Physics on the Plains 2022 Part 2	Oct. 2022
XV International Conference on Interconnections between Particle Physics and Cosmology	June 2022
Particle Physics on the Plains 2022	April 2022

RESEARCH MENTORING

Nathan Kura (High School Student)	Sep. 2026 –
Roberto Caiozzo (Graduate, SISSA)	Nov. 2025 –
Aaroodd Ujjayini Ramachandran (Graduate, Washington University in St. Louis)	May 2024 –
Robert Ellis (Undergraduate, Washington University in St. Louis)	May 2024 –

TEACHING AND ACADEMIC SERVICE

Organizer, IFPU Seminar, SISSA	Sep. 2026 – present
Organizer, SISSA–ICTP Joint Journal Club	Sep. 2026 – present
Organizer, Theory Seminar, Washington University in St. Louis	Aug. 2023 – July 2024

Organizer, Student/Postdoc Q&A Sessions in Theory Seminar, Washington University in St. Louis — Aug. 2022 – July 2024

Teaching Assistant, Washington University in St. Louis

Group Theory and Symmetries in Physics (PHY 543)	Spring 2023
Classical Electrodynamics I (PHY 505)	Spring 2022
Mechanics (PHY 411)	Fall 2021

OUTREACH AND BROADER IMPACT

Outreach committee, Washington University in St. Louis — May 2022 – Aug. 2025

- A Summer Camp at the Claver House — June 2022
  Demonstrate and provide hands-on activities for physical phenomena related to angular momentum
- A Summer Camp at the Claver House — July 2022
  Demonstrate and provide hands-on activities for physical phenomena regarding the physics of waves